

Comparing Proactive and Reactive Robot Assistance in a Quiz-Based Interaction with Furhat

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ABSTRACT

This project investigates how the Furhat social robot can assist users during a timed quiz game by comparing two modes of support: reactive help, provided only when the user asks, and proactive help, triggered by gaze cues and periods of inactivity. In a between-subjects study with twenty participants, we measured enjoyment, sociability, usefulness and performance trust using established Human-Robot Interaction (HRI) questionnaires. Although we expected proactive assistance to create a more helpful and engaging interaction, the results did not support this hypothesis. Across all measures, the reactive mode received equal or higher scores and sociability showed a significant preference for reactive assistance. Qualitative comments further suggest that poorly timed proactive interventions and missed speech inputs reduced the perceived quality of the interaction. These findings highlight that proactive robot behaviour is not inherently better; its effectiveness depends strongly on a good implementation. Improving this may make proactive assistance more natural and supportive in future systems.

KEYWORDS

Social robotics, Furhat, quiz assistant, proactive help, reactive help, large language models, gaze

1 INTRODUCTION

Social robots are increasingly used to provide assistance in various fields, from education to healthcare. The potential for robots to support humans by teaching, guiding or providing information has been explored extensively, and today systems such as ChatGPT and Gemini are widely used as on-demand help tools. We wanted to explore this potential further, but rather than focusing only on situations where the user actively asks for help (reactive assistance), we were interested in whether a robot could also offer help proactively.

To investigate this, we chose Furhat, a physically embodied social robot capable of speech, gaze behaviour and expressive facial animations and used it as the host of a quiz-style game inspired by games such as *Who Wants to Be a Millionaire* [1]. The motivation for using a game-based scenario is that it places the user in a cognitively demanding situation where help may naturally be needed, making it possible to observe how assistance is perceived under pressure. This allows us to evaluate how the user’s perception of the robot’s usefulness and likeability changes between two modes of support: a reactive mode, in which help is given only upon request, and a proactive mode, in which the robot initiates help based on social cues such as the user’s inactivity or the user’s gaze.



Figure 1: Furhat used as the quiz game assistant

Ultimately, this project investigates how the Furhat social robot can support users during a timed quiz task by comparing these two assistance strategies.

2 RELATED WORK

Games have for long been recognised as an effective domain for studying HRI, as they allow researchers to analyse social behaviour, engagement and collaboration in a controlled yet motivating setting. Rato et al. argue that game-based environments offer a natural setting for studying interaction patterns and user experience [2]. This insight motivated the use of a quiz-game scenario for our project, in which the robot acts as both a host and an assistant. In a related context, Tutul and Pinkwart developed a multimodal robot quiz system, demonstrating that timing, fairness mechanisms and feedback modalities substantially influence user performance and perceived enjoyment [3].

Another major component of our study is the comparison between proactive and reactive robot assistance. Keidar et al. investigated these interaction modes in a mobile telecare robot and found that proactive behaviour can improve support but also risks being perceived as intrusive when delivered at inappropriate moments [4]. This motivates our decision to refine proactive behaviour using social cues, such as gaze, that can improve its timing and naturalness.

Gaze has been widely studied as an informative social cue in HRI. Nicora et al. showed that gaze detection is an effective trigger for initiating collaboration, enabling robots to determine when users require assistance or attention [5]. Incorporating gaze into our proactive mode builds directly on this insight, allowing the robot to offer help at moments that better align with the user’s attentional state.

Finally, recent progress in Large Language Models (LLMs) has introduced new opportunities for adaptive, context-aware robot behaviour. Kim, Lee and Mutlu provide an overview of LLM-powered

HRI, emphasising both the potential benefits and the methodological challenges associated with deploying LLMs in interactive robotic systems [6]. Our system draws on this line of work by using an LLM not only to generate contextually relevant hints but also to select facial expressions for the robot, enabling richer multimodal interaction.

In summary, although prior research has examined game-based HRI, proactive and reactive robot assistance, gaze-triggered support and LLM-driven robot behaviour, these elements have been studied independently. Our work addresses a gap in the literature by integrating these components within a single system.

3 METHOD

3.1 Research Question and Hypothesis

Research question: Is there a difference in user experience between proactive and reactive interaction modes of the Furhat robot during a quiz game?

Based on prior work showing that proactive social robots can increase engagement, we hypothesise that:

H1: A proactive Furhat robot will be perceived as more enjoyable, useful, sociable and trustworthy in its performance than the reactive robot during a quiz game.

The null hypothesis states that no differences will be found between the conditions on our dependent measures.

3.2 Measures

The independent variable in this study is the interaction mode of Furhat: *reactive* or *proactive*.

The dependent variables are *perceived enjoyability*, *perceived usefulness*, *perceived sociability* and *performance trust* assessed through established HRI questionnaires.

Perceived enjoyability, perceived usefulness and perceived sociability are measured using items derived from the Almere Model. Participants rate their agreement with each statement on a 5-point Likert scale ranging from “Strongly Disagree” to “Strongly Agree,” following Heerink et al. [7].

Performance trust is measured using the Multi-Dimensional Model of Trust (MDMT) framework [8][9][10][11][12]. Participants rate competence-related attributes (e.g., capable, skilled, competent, meticulous) and reliability-related attributes (e.g., predictable, reliable, consistent, dependable) on a 0–7 Likert scale, where 0 indicates “not at all” and 7 indicates “very.”

These measures capture participants’ subjective experience of the interaction, allowing us to evaluate how the two assistance modes influence enjoyment, perceived utility, sociability and trust.

3.3 Experimental Design

A between-subjects design with one independent variable, *interaction mode* (*reactive* vs. *proactive*) is used. Each participant was assigned randomly to one of these modes.

Each participant completes a single quiz session with eight multiple-choice questions (four medium, four hard). Questions are stored in a JSON database and retrieved via the `QuestionDatabase` class, which samples without replacement per difficulty to avoid repetition within a session. The quiz is presented on a laptop-based

web interface served by a Flask application (`QuizWebServer`), while Furhat provides speech, gaze and facial expressions.

In both conditions, the overall content (number of questions, difficulty structure, time limit of 60 s per question) is identical. The only systematic difference is whether help is provided only on explicit request (reactive) or can also be offered proactively based on inactivity and gaze (proactive), which isolates the effect of interaction mode on user experience.

3.4 Procedure

Each session follows a fixed sequence:

- (1) **Introduction and consent:** The experimenter welcomes the participant, explains the purpose of the study and obtains informed consent.
- (2) **Pre-study questionnaire:** Once seated, the participant completes the pre-study questionnaire assessing demographics and prior experience with robots.
- (3) **Instructions and setup:** After the questionnaire, the experimenter provides further instructions on how to interact with Furhat and the quiz interface. The participant is positioned in front of Furhat and the laptop.
- (4) **Start of the experiment:** The experimenter launches the Flask server and the web interface. When the participant presses “Tap to start” in the browser, the Experiment-Controller records the assigned interaction mode and initiates the quiz.
- (5) **Question loop (8 questions):**
 - (a) The controller draws a question of the appropriate difficulty from the database and calls `display_question()` on the GUI, which shows the text, four options and a 60 s timer.
 - (b) Furhat verbally introduces the question. In the proactive condition, a `ProactiveHelpStateMachine` is started with an inactivity timeout.
 - (c) During the answer window, the participant may request help (reactive/proactive) or respond to help offers (only proactive). User speech is sent to `LLMAPI`, which generates short responses or hints without revealing the correct answer.
 - (d) The participant selects an option or the timer expires. The GUI sends the result to the controller, which logs correctness, gives immediate “Correct”/“Wrong” feedback with an expression and inserts a short pause between questions.
- (6) **Post-questionnaire and debrief:** After the last question, Furhat thanks the participant and the GUI signals completion. Participants then fill in the user-experience questionnaire and are debriefed about the purpose of the two interaction modes.

This procedure ensures that all participants receive the same quiz structure and timing, while differing only in how and when the robot offers help, directly targeting our research question.

3.5 Technical Implementation

The code for the robot and quiz was implemented as a multithreaded Python controller which manages the interactions between the participant, robot, graphical user interface and large language model.

System Architecture.

The system architecture is made up of four main components:

- **GUI:** Presents quiz questions, records answer selections and handles experiment phases (welcome, start signal, question readiness).
- **Furhat interface:** Handles robot speech, facial expressions, automatic speech recognition (ASR) and gaze/attention events.
- **Question Database:** Contains questions sorted by difficulty, with associated answers and a pre-made hints list.
- **LLM module:** Interprets user speech and generates context-aware conversational responses plus a recommended facial-expression label (from a predefined set).

All modules communicate via the central ExperimentController, which cycles through the questions during game-play.

Proactive mode state machine.

In the proactive mode, a finite-state machine (ProactiveHelpStateMachine) decides at what point the robot offers help. The machine consists of three states: WAIT, OFFER_HINT, and GIVE_HINT. Hints are triggered by:

- Timed inactivity (default 18 s)
- Detected gaze towards the robot
- User made requests (“help”, “hint”)

Each state has its own timers (e.g., 5s response timeout after offering help), which is cancelled on state exit. The machine pauses automatically while the robot is answering a user-initiated question.

Large Language Model.

The system uses the Mistral 7B chat model (accessed via the Mistral API as open-mistral-7b) to interpret user speech and generate short spoken replies together with a facial-expression label from a predefined set. To avoid line overflow in narrow columns, we typeset the model name as open-mistral-7b.

Each user utterance, together with the current quiz context (question and answer options), is first sent to an interpretation prompt that classifies it into one of five categories: *hint*, *fact*, *answer*, *greeting*, or *irrelevant*. This classification determines the subsequent behaviour:

- **Hint:** Treated as a request for help. No additional LLM call is made; instead, the controller selects the next pre-written hint from the question’s hints list.
- **Fact:** A factual question about the quiz topic. A second prompt asks the LLM for a concise answer (max. 20 words) that does not reveal the correct alternative, together with an expression label.
- **Answer:** The input would reveal the correct quiz answer. Furhat replies with a fixed refusal and a “concerned” expression—no further LLM call is made.
- **Greeting:** Social greetings or thanks. A dedicated greeting prompt elicits a short friendly reply and expression label.
- **Irrelevant:** Unrelated to the quiz. Furhat does not respond; the event is only logged.

This pipeline enables context-aware conversational behaviour while maintaining control over hint content and preventing the model from disclosing correct answers.

4 RESULTS

4.1 Participants demographics

Participant demographics were comparable across conditions, with similar age, gender distribution, robot familiarity and prior robot interaction experience.

The demographics are summarised in Table 1 and 2.

Table 1: Continuous demographics

	Reactive	Proactive	Overall
Age (years) mean (SD)	23.2 (2.3)	22.8 (1.3)	23.0 (1.8)
Robot familiarity (1–7)			
mean (SD)	3.7 (1.3)	3.9 (1.7)	3.8 (1.5)

Table 2: Categorical demographics

	Reactive	Proactive	Overall
Participants (n)	10	10	20
Gender – Female	5	5	10
Gender – Male	5	5	10
Prior robot interaction – No	4	5	9
Prior robot interaction – Yes	6	5	11

To address the research question, we compared user experience scores between the reactive and proactive conditions across four categories: Enjoyment, Usefulness, Sociability and Performance Trust. Table 3 presents the descriptive statistics for each condition.

	Enjoyment	Usefulness	Sociability	Perf. Trust
Reactive	3.70 (0.44)	3.43 (0.74)	3.50 (0.55)	5.08 (0.49)
Proactive	3.46 (0.74)	3.13 (0.71)	2.88 (0.66)	4.53 (0.90)

Table 3: Mean (SD) scores by condition. The first three measures use a 1–5 Likert scale; Trust uses a 1–7 Likert scale.

Across all categories, the descriptive results consistently show slightly higher mean scores in the reactive condition. However, it is necessary to determine whether these differences reflect statistically reliable effects. Before conducting inferential tests, we verified that the dependent variables did not violate normality using Shapiro–Wilk tests (all $p > .05$), supporting the use of independent-samples t -tests. The resulting test statistics are summarised in Table 4.

Scale	t	df	p-value	Cohen’s d
Enjoyment	0.88	14.72	0.395	0.392
Usefulness	0.93	17.97	0.365	0.415
Sociability	2.30	17.47	0.034	1.028
Perf. Trust	1.70	13.93	0.111	0.761

Table 4: Independent-samples t -tests comparing proactive and reactive conditions.

Figure 2 provides a visual overview of the score distributions.

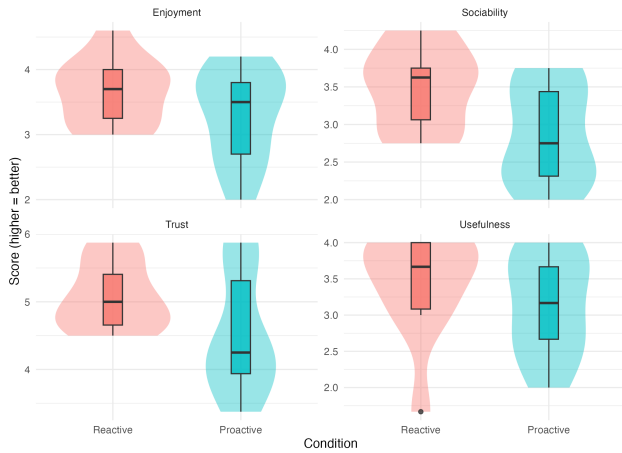


Figure 2: User-experience ratings by condition. Enjoyment, Sociability and Usefulness use a 1–5 scale; Trust uses a 1–7 scale.

Enjoyment

The reactive condition has a slightly higher mean enjoyment score (3.70) compared to the proactive condition (3.46). The reactive distribution is more compact, indicating more consistent enjoyment ratings, whereas the proactive scores display larger variability with several participants giving relatively low ratings. The t-test result ($p = 0.395$) shows no significant difference between conditions.

Usefulness

The reactive condition again shows a small advantage (3.43) compared to the proactive condition (3.13). The proactive condition shows broader variability and a more pronounced lower tail. The t-test result ($p = 0.365$) confirms again there is no significant difference.

Sociability

The reactive condition again scores higher on average (3.50) than the proactive condition (2.88). The proactive ratings show greater dispersion and several lower scores. Importantly, the updated t-test finds a significant difference ($p = 0.034$), with a large effect size ($d \approx 1.03$).

Performance Trust

The reactive condition shows higher trust scores on average (5.08 vs. 4.53). The proactive distribution is wider, including several lower trust ratings. The t-test result ($p = 0.111$) once again indicates no significant difference.

5 DISCUSSION

Although we hypothesised that proactive support—triggered through gaze cues and inactivity—would enhance enjoyment, usefulness, sociability and performance trust, the results did not provide evidence for this. Instead, the pattern of findings suggests that the proactive mode introduced interaction challenges that negatively influenced the user experience.

5.1 User Experience Measures

Participants did not report higher enjoyment or usefulness in the proactive condition. Qualitative comments indicate that proactive behaviour was sometimes poorly timed: users noted that Furhat interrupted them, misinterpreted their questions, didn’t hear them or occasionally provided too much information. Such disruptions likely reduced the perceived value of proactive help. Prior work highlights that proactive behaviours can feel intrusive when delivered at inappropriate moments, and our results are consistent with this insight.

Sociability was the only measure showing a significant difference between conditions, with the reactive robot perceived as more sociable. This outcome suggests that sociability depends strongly on timing. When proactive help was triggered unexpectedly, participants appeared to view the robot as less aware of the flow of the interaction.

Performance trust did not significantly differ between conditions, although trust ratings were slightly higher for the reactive robot. Reports of interruptions and incorrect or overly revealing hints in the proactive mode likely weakened perceptions of reliability and competence, the two pillars of performance trust in the MDMT framework.

5.2 Limitations and Future Work

Several limitations may have influenced the results. Technical issues affected both conditions: Furhat’s ASR frequently failed to detect or correctly interpret user speech, leading to missed or inappropriate responses. In addition, LLM-generated replies occasionally revealed the correct answer instead of providing neutral guidance.

The study was also constrained by a small sample size, which limits the statistical power and generalisability of the findings.

Future work should prioritise improving speech recognition, refining the timing and conditions for proactive help, and exploring tasks with lower time pressure. Recruiting a larger and more diverse participant pool would further strengthen the results.

Overall, proactive support was not perceived as more helpful or engaging. Its effectiveness appears to depend heavily on accurate sensing and well-timed interventions—areas where current system limitations favoured the smoother and more predictable reactive assistance.

6 CONCLUSION

This study compared proactive and reactive robot assistance during a timed quiz interaction with Furhat. Contrary to our expectations, proactive help did not lead to higher enjoyment, usefulness, sociability or performance trust. Instead, interruptions, mistimed offers of help and missed speech inputs reduced the perceived quality of the interaction, whereas the reactive mode was experienced as more socially appropriate.

These findings show that proactive support is not inherently beneficial; its success depends strongly on accurate timing, good performance and the robot’s ability to recognise when help is truly welcome. Improving gaze detection, refining inactivity thresholds and enhancing ASR may enable more natural proactive behaviour in the future. For now, well-timed, user-initiated assistance provides

a smoother and more trusted interaction, highlighting the need for proactive strategies that better respect human conversational flow.

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A CONTRIBUTION STATEMENT

Individual contributions were as follows:

- **Maria Sebastião:** Developed the GUI and ExperimentController; contributed to system integration; assisted with data processing.
- **Diogo Paulo:** Developed the GUI and ExperimentController; contributed to system integration; conducted user studies; assisted with data processing.
- **Hugo Dezerto:** Developed the Furhat API components; contributed to the ExperimentController and overall system integration.
- **Cilia Nanninga:** Created the question database and user questionnaires; conducted user studies; performed data analysis in R.
- **Helena Westermann:** Developed the LLM interface and contributed to the system architecture; conducted user studies.

B CONSENT FORM

Purpose of the Study: You are invited to participate in a user study as part of the KTH course DD2413 Social Robotics. The purpose of the study is to investigate how users perceive interactions with a Furhat social robot during a short quiz task.

What You Will Do: You will interact with the Furhat robot, answer a series of quiz questions, and then complete a short questionnaire about your experience. The study takes approximately 5–10 minutes.

Voluntary Participation: Your participation is entirely voluntary. You may withdraw at any time without giving a reason and without any consequences.

Data Handling: Your responses will be collected anonymously. No personal data that can identify you will be stored. The data will be used only for educational and research purposes within the course.

Risks and Benefits: There are no expected risks associated with this study. While there may be no direct personal benefits, your participation will contribute to research and education in social robotics.

C QUESTIONNAIRES

Table 5: Pre-study questionnaire items used in the user study.

Q	Item
1	I have read and agree to the consent form. (Yes/No)
2	How old are you? (open numerical input)
3	What is your gender? (Non-binary / Female / Male / Prefer not to say / Other)
4	Have you interacted with robots before? (Yes/No)
5	How familiar are you with robots in general? (1 = Not at all, 7 = Very familiar)

Table 6: Post-study questionnaire items used in the user study.

Q	Item
Section 1 — Almere Model (1–5 Likert scale)	
1	I think the robot is useful to me.
2	I find the robot boring.
3	I feel the robot understands me.
4	I enjoy doing things with the robot.
5	I find the robot pleasant to interact with.
6	It would be convenient for me to have the robot.
7	I think the robot can help me with many things.
8	I consider the robot a pleasant conversational partner.
9	I find the robot fascinating.
10	I enjoy the robot talking to me.
11	I think the robot is nice.
12	I find the robot enjoyable.
Section 2 — MDMT (1–7 scale)	
13	Capable.
14	Predictable.
15	Meticulous.

Q	Item (continued)
16	Reliable.
17	Skilled.
18	Competent.
19	Consistent.
20	Dependable.
Section 3 – Open Comment	
21	Do you have any comments on the interaction with the robot? (open response)

D USER COMMENTS

Table 7: User comments in the reactive interaction mode

	Comment
1	Sometimes the robot gives the answer to the question without being asked to do it.
2	It is nice to interact with the robot; the robot could give more hints.
3	I enjoy his facial expressions.
4	He didn't seem to recognize when I was asking a question vs. talking to myself; sometimes he jumped to giving the answer rather than a hint.
5	It felt reassuring when it did a nod gesture.
6	Sometimes frustrating.
7	It was a little weird to interact with.
8	He gives a hint but doesn't react to specific questions.
9	It didn't answer my last question.
10	A bit monotonous; some more fun during the conversation would be cool.

Table 8: User comments in the proactive interaction mode

	Comment
1	The conversation went smoothly and the impression of the robot was quite good, although there were some interruptions from the robot while I was trying to speak.
2	No comment.
3	Sometimes the smile was creepy.
4	Not really; I was very impressed. He sometimes missed or needed clarification that I needed help.
5	He didn't understand many of my questions and did not respond, so I wasn't sure if he had heard me.
6	It's a bit weird.
7	One time he gave an answer even though he just wanted to give a hint.
8	Sometimes the robot interrupted me or offered help when I didn't ask for it.
9	It gave one answer away when it just wanted to give a hint (the Plato question).
10	He sometimes did not hear me.

E FLOWCHART OF THE PROCEDURE

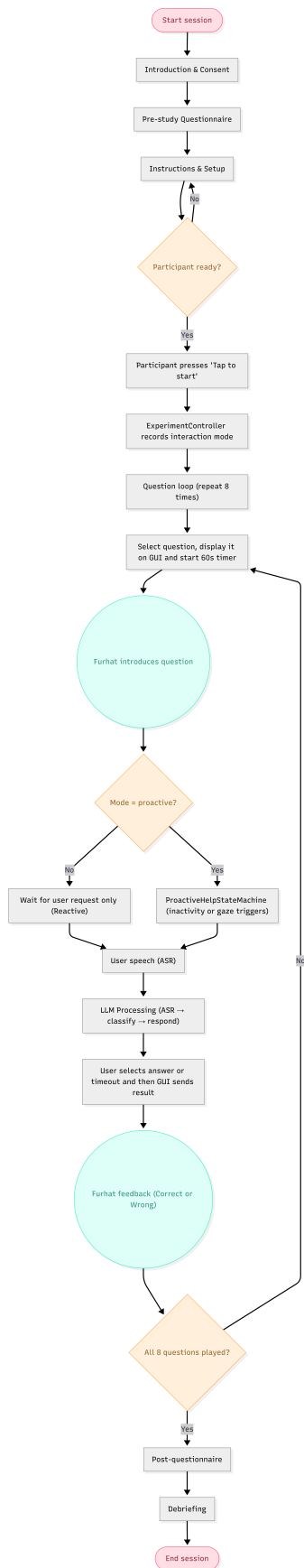


Figure 3: Procedure flowchart